

California Education & Income Analysis

Seven years of population data across age, gender, education, and personal income — explored with joins, window functions, CTEs, and subqueries.

1,060 records

7 years

5 education levels

8 income bands

MySQL 8.0+ Advanced Analytics Project

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What the data covers, and what we set out to find

The dataset tracks California's population by Age group, Gender, Educational Attainment, and Personal Income band, for every year from 2008 to 2014. The goal was to move beyond a single table of numbers and use SQL (joins, window functions, CTEs, and subqueries) to uncover how education and income shifted across the recession-recovery period.

1,060

Total records

7

Years (2008–14)

3

Age groups

2

Genders

5

Education levels

8

Income bands

Guiding questions

- How did total population, and the male–female balance, evolve 2008–14?
- Which education levels and income bands grew the fastest — and which shrank?
- Which demographic segments gained the most people, and which declined?

SQL techniques used to answer 28 analytical questions

1

Self Joins

Paired each segment's 2008 row against its 2014 row to measure growth directly, combo by combo.

2

Window Functions

RANK, DENSE_RANK, ROW_NUMBER and LAG to rank categories and track year-over-year change.

3

CTEs

Layered WITH clauses to build yearly aggregates, then classify, compare, and filter on top of them.

4

Subqueries

Correlated and nested subqueries to find above-average, top, and second-highest segments.

Project scope & architecture

A comprehensive multi-dimensional analysis using the **ca_education_income** table.

Core Attributes

Dimensions cover Year, Age, Gender, Educational Attainment, and Personal Income against Population Count.

28 Business Questions

Progressive analytical stack testing complex joins, window sequences, modular CTEs, and nested subqueries.

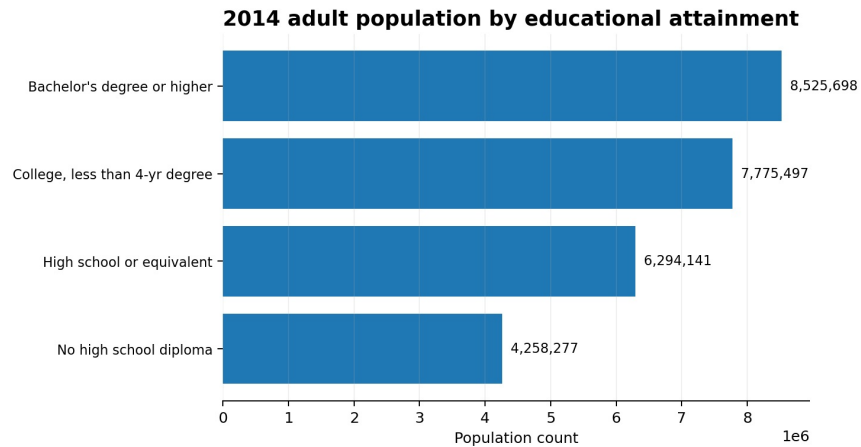
Time Horizon

Longitudinal tracking of California socioeconomic movements from 2008 through 2014.

The data at a glance

2014 adult population by educational attainment

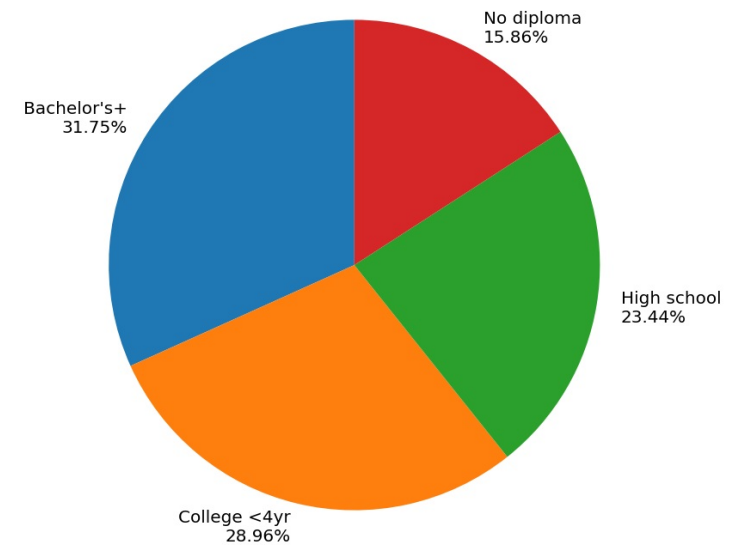
Bachelor's degree holders are the single largest group among California adults.



2014 adult population share by education

Bachelor's+ and some-college adults together make up over 60% of the adult population.

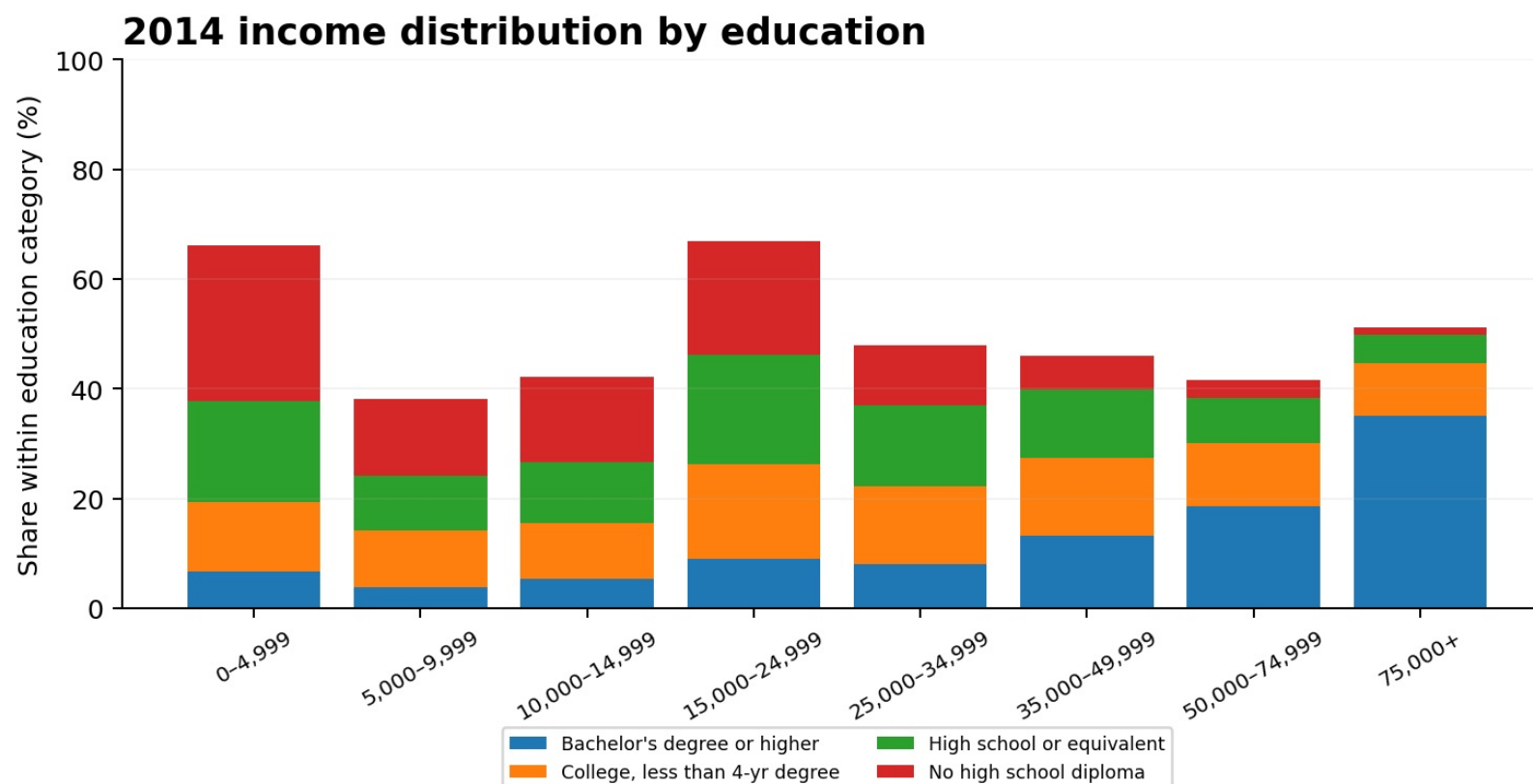
2014 adult population share by education



Income mix, by education

2014 income distribution by education

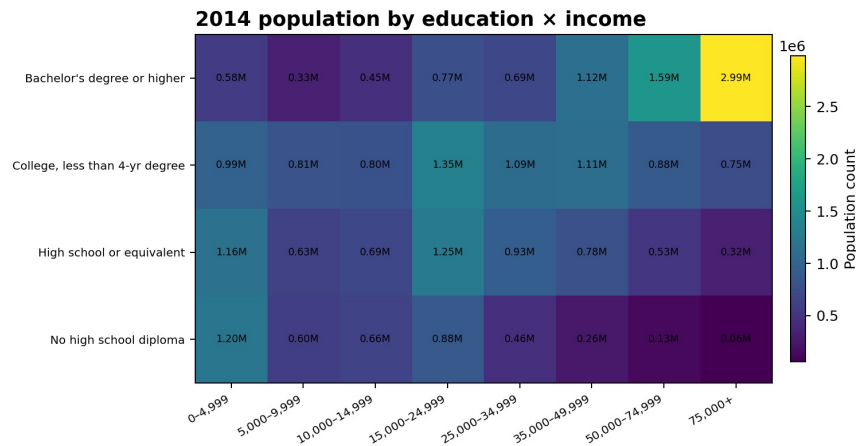
Each income band's mix of educational attainment — higher-income bands skew toward bachelor's degree holders.



Education × income, side by side

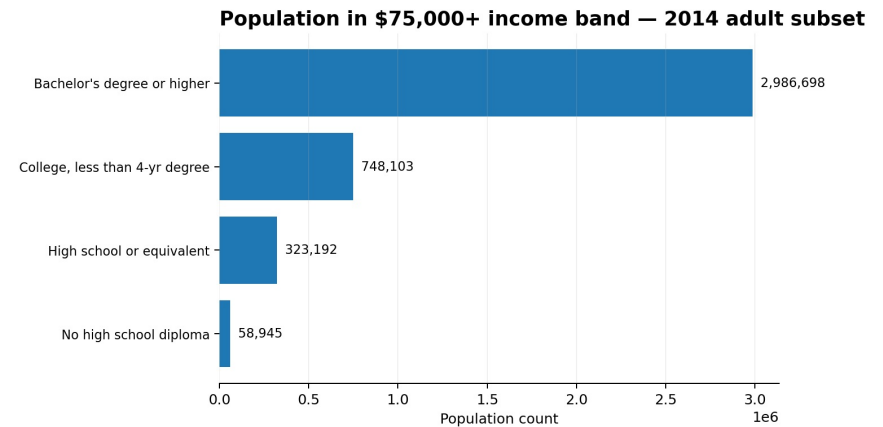
2014 population by education × income

Bachelor's degree holders dominate the top income band (\$75,000+).



Population in the \$75,000+ income band

Nearly 3M bachelor's degree holders sit in the top income band — far more than any other group.



SQL Analysis Questions 1–2

- 1 Compare 2008 and 2014 population for every combination of Age, Gender, Educational Attainment and Personal Income. Show population in both years and the change.

```

3  -- in both years and the change.
4
5  • select a.Age, a.Gender, a.`Educational Attainment`, a.`Personal Income`,
6      a.`Population Count` as population_2008, b.`Population Count` as population_2014,
7      b.`Population Count` - a.`Population Count` as population_change
8  from edu_inc a join edu_inc b
9      on a.Age = b.Age
10     and a.Gender = b.Gender
11     and a.`Educational Attainment` = b.`Educational Attainment`
12     and a.`Personal Income` = b.`Personal Income`
13 where a.Year = '01-01-2008' and b.Year = '01-01-2014';
14
15

```

Age	Gender	Educational Attainment	Personal Income	population_2008	population_2014	population_change
00 to 17	Male	Children under 15	No Income	0	0	0
00 to 17	Male	No high school diploma	No Income	650889	610313	-40576
00 to 17	Male	No high school diploma	\$5,000 to \$9,999	30152	26380	-3772
00 to 17	Male	No high school diploma	\$10,000 to \$14,999	7092	11857	4765
00 to 17	Male	No high school diploma	\$15,000 to \$24,999	3974	3920	-54

Result 12 x

Read Only

- 2 Find demographic segments whose population increased by more than 10,000 people between 2008 and 2014. Use a SELF JOIN.

```

18
19 • select a.Age, a.Gender, a.`Educational Attainment`, a.`Personal Income`,
20     a.`Population Count` as population_2008, b.`Population Count` as population_2014,
21     b.`Population Count` - a.`Population Count` as population_change
22 from edu_inc a join edu_inc b
23     on a.Age = b.Age
24     and a.Gender = b.Gender
25     and a.`Educational Attainment` = b.`Educational Attainment`
26     and a.`Personal Income` = b.`Personal Income`
27 where a.Year = '01-01-2008'
28     and b.Year = '01-01-2014'
29     and b.`Population Count` - a.`Population Count` > 10000;
30

```

Age	Gender	Educational Attainment	Personal Income	population_2008	population_2014	population_change
00 to 17	Male	High school or equivalent	No Income	0	15741	15741
18 to 64	Male	No high school diploma	No Income	261172	376783	115611
18 to 64	Male	No high school diploma	\$75,000 and over	44201	54734	10443
18 to 64	Male	High school or equivalent	No Income	296555	405204	108649
18 to 64	Male	High school or equivalent	\$5,000 to \$9,999	187833	238754	50921

Result 13 x

Read Only

SQL Analysis Questions 3–4

- 3 Compare the population for each Gender + Educational Attainment combination between 2008 and 2014, aggregating across Age and Personal Income.

```

34
35 • select a.Gender, a.`Educational Attainment`,
36        sum(a.`Population Count`) as population_2008,
37        sum(b.`Population Count`) as population_2014,
38        sum(b.`Population Count`) - sum(a.`Population Count`) as population_change
39 from edu_inc a join edu_inc b
40 on a.Gender = b.Gender
41    and a.`Educational Attainment` = b.`Educational Attainment`
42    and a.Age = b.Age
43    and a.`Personal Income` = b.`Personal Income`
44 where a.Year = '01-01-2008'
45        and b.Year = '01-01-2014'
46 group by a.Gender, a.`Educational Attainment`;

```

Gender	Educational Attainment	population_2008	population_2014	population_change
Male	Children under 15	0	0	0
Male	No high school diploma	3073168	2811001	-262167
Male	High school or equivalent	3081357	3294850	213493
Male	Some college, less than 4-yr degree	3487582	3685424	197842
Female	Children under 15	0	0	0

Result 14 x

Read Only

- 4 Compare Male and Female population for each Educational Attainment and Personal Income combination in 2014. Show the gender difference.

```

51
52 • select a.`Educational Attainment`, a.`Personal Income`,
53        a.`Population Count` as male_population,
54        b.`Population Count` as female_population,
55        a.`Population Count` - b.`Population Count` as gender_difference
56 from edu_inc a join edu_inc b
57 on a.Age = b.Age
58    and a.`Educational Attainment` = b.`Educational Attainment`
59    and a.`Personal Income` = b.`Personal Income`
60 where a.Year = '01-01-2014'
61        and b.Year = '01-01-2014'
62        and a.Gender = 'Male'
63        and b.Gender = 'Female';

```

Gender	Educational Attainment	population_2008	population_2014	population_change
Male	Children under 15	0	0	0
Male	No high school diploma	3073168	2811001	-262167
Male	High school or equivalent	3081357	3294850	213493
Male	Some college, less than 4-yr degree	3487582	3685424	197842
Female	Children under 15	0	0	0

Result 14 x

Read Only

SQL Analysis Questions 5–6

- 5 Compare 2013 and 2014 population for every Age + Gender segment, aggregating across Education and Income. Find segments that declined.

```

69 • select a.Age, a.Gender,
70         sum(a.`Population Count`) as population_2013,
71         sum(b.`Population Count`) as population_2014,
72         sum(b.`Population Count`) - sum(a.`Population Count`) as population_change
73 from edu_inc a join edu_inc b
74 on a.Age = b.Age
75 and a.Gender = b.Gender
76 and a.`Educational Attainment` = b.`Educational Attainment`
77 and a.`Personal Income` = b.`Personal Income`
78 where a.Year = '01-01-2013'
79 and b.Year = '01-01-2014'
80 group by a.Age, a.Gender
81 having sum(b.`Population Count`) < sum(a.`Population Count`)

```

Age	Gender	population_2013	population_2014	population_change
00 to 17	Male	688204	670259	-17945
00 to 17	Female	707295	675989	-31306

Result 15 x

Read Only

- 6 For each Personal Income band, compare total population in 2008 versus 2014, aggregating across Age, Gender and Education.

```

87 • select a.`Personal Income`,
88         sum(a.`Population Count`) as population_2008,
89         sum(b.`Population Count`) as population_2014,
90         sum(b.`Population Count`) - sum(a.`Population Count`) as population_change
91 from edu_inc a join edu_inc b
92 on a.`Personal Income` = b.`Personal Income`
93 and a.Age = b.Age
94 and a.Gender = b.Gender
95 and a.`Educational Attainment` = b.`Educational Attainment`
96 where a.Year = '01-01-2008'
97 and b.Year = '01-01-2014'
98 group by a.`Personal Income`;

```

Personal Income	population_2008	population_2014	population_change
No Income	4331910	5192554	860644
\$5,000 to \$9,999	2317874	2405313	87439
\$10,000 to \$14,999	2652090	2633938	-18152
\$15,000 to \$24,999	4036167	4252400	216233
\$25,000 to \$34,999	3003788	3179308	175520
\$35,000 to \$49,999	3319746	3278595	-41151
\$50,000 to \$74,999	3354245	3136540	-217705

Result 16 x

Read Only

7 Rank Educational Attainment categories by total population within each year using RANK().

```

104 • select Year, `Educational Attainment`,
105         sum(`Population Count`) as total_population,
106         rank() over(partition by Year order by sum(`Population Count`) desc) as population_rank
107 from edu_inc
108 group by Year, `Educational Attainment`;
109
110
111

```

Year	Educational Attainment	total_population	population_rank
01-01-2008	Bachelor's degree or higher	7372964	1
01-01-2008	Some college, less than 4-yr degree	7236983	2
01-01-2008	No high school diploma	5964588	3
01-01-2008	High school or equivalent	5957715	4
01-01-2008	Children under 15	0	5
01-01-2009	Bachelor's degree or higher	7436401	1
01-01-2009	Some college, less than 4-yr degree	7255063	2
01-01-2009	No high school diploma	5964588	3
01-01-2009	High school or equivalent	5957715	4
01-01-2009	Children under 15	0	5

8 Rank Personal Income bands by population within each year using DENSE_RANK().

```

117 • select Year, `Personal Income`,
118         sum(`Population Count`) as total_population,
119         dense_rank() over(partition by Year order by sum(`Population Count`) desc) as population_rank
120 from edu_inc
121 group by Year, `Personal Income`;
122
123
124

```

Year	Personal Income	total_population	population_rank
01-01-2008	No Income	4340574	1
01-01-2008	\$15,000 to \$24,999	4038176	2
01-01-2008	\$75,000 and over	3499620	3
01-01-2008	\$50,000 to \$74,999	3354245	4
01-01-2008	\$35,000 to \$49,999	3321973	5
01-01-2008	\$25,000 to \$34,999	3006394	6
01-01-2008	\$10,000 to \$14,999	2653394	7
01-01-2008	\$5,000 to \$9,999	2237874	8
01-01-2008	\$0 to \$4,999	1999999	9

SQL Analysis Questions 9–10

- 9 Find the top 3 Educational Attainment categories by population for every year using ROW_NUMBER().

```

126
127 • select Year, `Educational Attainment`, total_population
128 from (
129     select Year, `Educational Attainment`,
130         sum(`Population Count`) as total_population,
131         row_number() over(partition by Year order by sum(`Population Count`) desc) as rn
132     from edu_inc
133     group by Year, `Educational Attainment`
134 ) a
135 where rn <= 3;

```

Year	Educational Attainment	total_population
01-01-2008	Bachelor's degree or higher	7172964
01-01-2008	Some college, less than 4-yr degree	7236983
01-01-2008	No high school diploma	5964588
01-01-2009	Bachelor's degree or higher	7436401
01-01-2009	Some college, less than 4-yr degree	7255063
01-01-2009	No high school diploma	5902137
01-01-2010	Bachelor's degree or higher	7648905

- 10 Calculate previous-year population for every Gender + Educational Attainment combination using LAG(), and calculate the population change.

```

141 • select Year, Gender, `Educational Attainment`, total_population,
142     lag(total_population) over(partition by Gender, `Educational Attainment` order by Year) as previous_population,
143     total_population - lag(total_population) over(partition by Gender, `Educational Attainment` order by Year)
144     as population_change
145 from (
146     select Year, Gender, `Educational Attainment`,
147         sum(`Population Count`) as total_population
148     from edu_inc
149     group by Year, Gender, `Educational Attainment`
150 ) aj;

```

Year	Gender	Educational Attainment	total_population	previous_population	population_change
01-01-2008	Female	Bachelor's degree or higher	3630986		
01-01-2009	Female	Bachelor's degree or higher	3587347	3630986	-43639
01-01-2010	Female	Bachelor's degree or higher	3753679	3587347	166332
01-01-2011	Female	Bachelor's degree or higher	3662573	3753679	-91106
01-01-2012	Female	Bachelor's degree or higher	3800850	3662573	138277
01-01-2013	Female	Bachelor's degree or higher	4011603	3800850	210753
01-01-2014	Female	Bachelor's degree or higher	4201713	4011603	190110
01-01-2008	Female	Children under 15	0		
01-01-2009	Female	Children under 15	0	0	0
01-01-2010	Female	Children under 15	0	0	0

11 Calculate year-over-year population growth percentage for every Age + Gender combination using LAG().

```

156 • select Year, Age, Gender, total_population,
157       lag(total_population) over(partition by Age, Gender order by Year) as previous_population,
158       round(((total_population - lag(total_population) over(partition by Age, Gender order by Year))
159             / lag(total_population) over(partition by Age, Gender order by Year) * 100, 2) as growth_percentage
160 from (
161     select Year, Age, Gender,
162            sum("Population Count") as total_population
163     from edu_inc
164     group by Year, Age, Gender
165 ) a;

```

Year	Age	Gender	total_population	previous_population	growth_percentage
01-01-2008	00 to 17	Female	698888		
01-01-2009	00 to 17	Female	712227	698888	1.91
01-01-2010	00 to 17	Female	663685	712227	-6.82
01-01-2011	00 to 17	Female	695767	663685	4.83
01-01-2012	00 to 17	Female	694577	695767	-0.17
01-01-2013	00 to 17	Female	715607	694577	3.03
01-01-2014	00 to 17	Female	685916	715607	-4.15
01-01-2008	00 to 17	Male	710373		
01-01-2009	00 to 17	Male	696875	710373	-1.90
01-01-2010	00 to 17	Male	644676	696875	-7.40

12 Calculate cumulative population for each Educational Attainment category from 2008 through 2014.

```

171 • select Year, `Educational Attainment`,
172       sum(total_population) over(partition by `Educational Attainment` order by Year) as cumulative_population
173 from (
174     select Year, `Educational Attainment`,
175            sum("Population Count") as total_population
176     from edu_inc
177     group by Year, `Educational Attainment`
178 ) a;

```

Year	Educational Attainment	cumulative_population
01-01-2008	Bachelor's degree or higher	7372964
01-01-2009	Bachelor's degree or higher	14809365
01-01-2010	Bachelor's degree or higher	22458270
01-01-2011	Bachelor's degree or higher	30022437
01-01-2012	Bachelor's degree or higher	37788270
01-01-2013	Bachelor's degree or higher	46089507
01-01-2014	Bachelor's degree or higher	54617676
01-01-2008	Children under 15	0
01-01-2009	Children under 15	0
01-01-2010	Children under 15	0

SQL Analysis Questions 13–14

- 13 Calculate each Personal Income band's percentage share of total population within each year.

```

184 • select Year, `Personal Income`,
185        sum(`Population Count`) as total_population,
186        round(sum(`Population Count`) /
187              sum(sum(`Population Count`)) over(partition by Year) * 100, 2) as population_share
188 from edu_inc
189 group by Year, `Personal Income`;
190
191

```

Year	Personal Income	total_population	population_share
01-01-2008	No Income	4340574	16.36
01-01-2008	\$5,000 to \$9,999	2317874	8.74
01-01-2008	\$10,000 to \$14,999	2653394	10.00
01-01-2008	\$15,000 to \$24,999	4038176	15.22
01-01-2008	\$25,000 to \$34,999	3006394	11.33
01-01-2008	\$35,000 to \$49,999	3321973	12.52
01-01-2008	\$50,000 to \$74,999	3354245	12.64
01-01-2008	\$75,000 and over	3499620	13.19
01-01-2009	No Income	4170709	15.77
01-01-2009	\$5,000 to \$9,999	2411487	9.11

Result 24 x

Output



- 14 For every year, identify the most populated Gender + Age segment using ROW_NUMBER().

```

194
195 • select Year, Gender, Age, total_population
196 from (
197     select Year, Gender, Age,
198            sum(`Population Count`) as total_population,
199            row_number() over(partition by Year order by sum(`Population Count`) desc) as rn
200     from edu_inc
201     group by Year, Gender, Age
202 ) a
203 where rn = 1;

```

Year	Gender	Age	total_population
01-01-2008	Male	18 to 64	11072116
01-01-2009	Male	18 to 64	11040468
01-01-2010	Male	18 to 64	10954078
01-01-2011	Male	18 to 64	11115121
01-01-2012	Male	18 to 64	11254245
01-01-2013	Male	18 to 64	11323361
01-01-2014	Male	18 to 64	11382669

Result 25 x

Output



15 Find the year in which each Educational Attainment category had its highest population.

```

208
209 • select `Educational Attainment`, Year, total_population
210 from (
211     select `Educational Attainment`, Year,
212         sum(`Population Count`) as total_population,
213         row_number() over(partition by `Educational Attainment` order by sum(`Population Count`) desc) as
214         from edu_inc
215         group by `Educational Attainment`, Year
216     ) a
217     where rn = 1;

```

Educational Attainment	Year	total_population
Bachelor's degree or higher	01-01-2014	8528169
Children under 15	01-01-2010	0
High school or equivalent	01-01-2012	6372967
No high school diploma	01-01-2012	6002290
Some college, less than 4-yr degree	01-01-2013	7850041

Result 26 x

Output

16 Compare each year's total population with the previous year using LAG(), and identify the largest annual increase.

```

223 • select Year, total_population,
224     lag(total_population) over(order by Year) as previous_population,
225     total_population - lag(total_population) over(order by Year) as population_change
226 from (
227     select Year, sum(`Population Count`) as total_population
228     from edu_inc
229     group by Year
230 ) a
231 order by population_change desc
232 limit 1;

```

Educational Attainment	Year	total_population
Bachelor's degree or higher	01-01-2014	8528169
Children under 15	01-01-2010	0
High school or equivalent	01-01-2012	6372967
No high school diploma	01-01-2012	6002290
Some college, less than 4-yr degree	01-01-2013	7850041

Result 26 x

Output

SQL Analysis Questions 17–18

- 17** Using CTEs, calculate total population by year and classify each year as High Population or Low Population based on the overall average.

```

238     select Year, sum('Population Count') as total_population
239     from edu_inc group by Year
240 ),
241 average_population as (
242     select avg(total_population) as avg_population
243     from yearly_population
244 )
245 select y.Year, y.total_population,
246        case
247          when y.total_population > a.avg_population then 'High Population' else 'Low Population'
248        end as population_category
249 from yearly_population y
250 cross join average_population a;

```

Year	total_population	population_category
01-01-2008	26532250	Low Population
01-01-2009	26442817	Low Population
01-01-2010	26469031	Low Population
01-01-2011	27160088	Low Population
01-01-2012	27641508	High Population
01-01-2013	27905466	High Population
01-01-2014	28711687	High Population

- 18** Using CTEs, calculate Educational Attainment population in 2008 and 2014 and return the top 5 categories by absolute change.

```

256 with education_population as (
257     select 'Educational Attainment',
258            sum(case when Year = '01-01-2008' then 'Population Count' else 0 end) as population_2008,
259            sum(case when Year = '01-01-2014' then 'Population Count' else 0 end) as population_2014
260     from edu_inc
261     group by 'Educational Attainment'
262 )
263 select 'Educational Attainment', population_2008, population_2014,
264        population_2014 - population_2008 as population_change,
265        abs(population_2014 - population_2008) as absolute_change
266 from education_population
267 order by absolute_change desc
268 limit 5;

```

Educational Attainment	population_2008	population_2014	population_change	absolute_change
Bachelor's degree or higher	7372964	8528169	1155205	1155205
Some college, less than 4-yr degree	7236983	7781293	544310	544310
No high school diploma	5964588	5583171	-381417	381417
High school or equivalent	5957715	6323174	365459	365459
Children under 15	0	0	0	0

SQL Analysis Questions 19–20

19 Using CTEs, calculate population by Gender and Year and find which gender had the larger population in each year.

```

274 • with gender_population as (
275     select Year, Gender, sum('Population Count') as total_population
276     from edu_inc
277     group by Year, Gender
278 )
279 select Year, Gender, total_population
280 from (
281     select Year, Gender, total_population,
282     row_number() over(partition by Year order by total_population desc) as rn
283     from gender_population
284 ) a
285 where rn = 1;
286

```

Year	Gender	total_population
01-01-2008	Male	13398886
01-01-2009	Male	13429050
01-01-2010	Male	13356399
01-01-2011	Male	13720944
01-01-2012	Male	13929862
01-01-2013	Male	14036096
01-01-2014	Male	14174174

Result 30 x

Output

20 Using CTEs, calculate average annual population for each Age group and identify the Age group with the highest average.

```

290
291 • with age_population as (
292     select Age, Year, sum('Population Count') as total_population
293     from edu_inc
294     group by Age, Year
295 )
296 select Age, avg(total_population) as average_population
297 from age_population
298 group by Age
299 order by average_population desc
300 limit 1;

```

Age	average_population
18 to 64	21686699.2857

Result 31 x

Output

SQL Analysis Questions 21–22

21 Using CTEs, identify Educational Attainment categories whose population increased in at least 4 of the 6 year-to-year transitions.

```

306 with education_population as (
307     select `Educational Attainment`, Year,
308           sum(`Population Count`) as total_population
309     from edu_inc
310     group by `Educational Attainment`, Year ),
311 changes as (
312     select `Educational Attainment`, Year, total_population,
313           lag(total_population) over(partition by `Educational Attainment` order by Year) as previous_population
314     from education_population)
315 select `Educational Attainment` from changes
316 group by `Educational Attainment`
317 having sum(case when total_population > previous_population then 1 else 0 end) >= 4;
318

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Educational Attainment
Bachelor's degree or higher
High school or equivalent
Some college, less than 4-yr degree

Result 32 x

Output



22 Using CTEs, calculate the average yearly population share of each Personal Income band and find bands whose average share exceeds 10%.

```

322
323 with income_share as (
324     select Year, `Personal Income`,
325           sum(`Population Count`) /
326           sum(sum(`Population Count`)) over(partition by Year) * 100 as population_share
327     from edu_inc
328     group by Year, `Personal Income`
329 )
330 select `Personal Income`, avg(population_share) as average_share
331 from income_share
332 group by `Personal Income`
333 having avg(population_share) > 10;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Personal Income	average_share
No Income	17.39878571
\$15,000 to \$24,999	15.46761429
\$25,000 to \$34,999	11.15538571
\$35,000 to \$49,999	11.76822857
\$50,000 to \$74,999	11.74317143
\$75,000 and over	13.66405714

Result 34 x

Output



SQL Analysis Questions 23–24

- 23 Find Educational Attainment categories whose 2014 population was greater than the average 2014 population across all education categories. Use a subquery.

```
139 • select `Educational Attainment`, sum(`Population Count`) as total_population
140 from edu_inc
141 where Year = '01-01-2014'
142 group by `Educational Attainment`
143 having sum(`Population Count`) > (
144     select avg(total_population)
145     from (
146         select sum(`Population Count`) as total_population
147         from edu_inc
148         where Year = '01-01-2014'
149         group by `Educational Attainment`
150     ) a );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Educational Attainment	total_population
High school or equivalent	6323174
Some college, less than 4-yr degree	7781293
Bachelor's degree or higher	8528169

Result 35 x

Output

- 24 Find Personal Income bands whose 2014 population was greater than the average income-band population in 2014. Use a subquery.

```
356 • select `Personal Income`, sum(`Population Count`) as total_population
357 from edu_inc
358 where Year = '01-01-2014'
359 group by `Personal Income`
360 having sum(`Population Count`) > (
361     select avg(total_population)
362     from (
363         select sum(`Population Count`) as total_population
364         from edu_inc
365         where Year = '01-01-2014'
366         group by `Personal Income`
367     ) a );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Personal Income	total_population
No Income	5192554
\$15,000 to \$24,999	4260763
\$75,000 and over	4116938

Result 36 x

Output

25 Find the most populated Age + Gender combination in 2014 using a subquery.

```

373 • select Age, Gender, sum(`Population Count`) as total_population
374 from edu_inc
375 where Year = '01-01-2014'
376 group by Age, Gender
377 having sum(`Population Count`) = (
378     select max(total_population)
379     from (
380         select sum(`Population Count`) as total_population
381         from edu_inc
382         where Year = '01-01-2014'
383         group by Age, Gender
384     ) a );

```

26 Find the second-highest Educational Attainment category by population in 2014 using subqueries. *No screenshot was available for this one — SQL reconstructed to match the style of the others.*

```

select `Educational Attainment`, sum(`Population Count`) as
total_population
from edu_inc
where Year = '01-01-2014'
group by `Educational Attainment`
having sum(`Population Count`) = (
    select max(total_population)
    from (
        select sum(`Population Count`) as total_population
        from edu_inc
        where Year = '01-01-2014'
        group by `Educational Attainment`
    ) a
    where total_population < (
        select max(total_population)
        from (
            select sum(`Population Count`) as
total_population
            from edu_inc
            where Year = '01-01-2014'
            group by `Educational Attainment`
        ) b
    )

```

Educational Attainment	Population 2014
Some college, less than 4-yr degree	7781293

SQL Analysis Questions 27–28

- 27 Find Age + Gender combinations whose average annual population from 2008–2014 is above the overall average annual population across all Age + Gender combinations.

```

422 -- all Age + Gender combinations.
423
424 select Age, Gender, avg(total_population) as average_population
425 from (
426     select Age, Gender, Year, sum('Population Count') as total_population
427     from edu_inc
428     group by Age, Gender, Year
429 ) a
430 group by Age, Gender
431 having avg(total_population) > (
432     select avg(total_population)
433     from (
434         select Age, Gender, Year, sum('Population Count') as total_population
435         from edu_inc
436         group by Age, Gender, Year
437     ) b );
438

```

Age	Gender	average_population
18 to 64	Male	11163151.1429
18 to 64	Female	10523548.1429

Result 38 x

Output

- 28 Find Educational Attainment categories whose 2014 population was higher than their own average population across 2008–2014.

```

443 sum(case when Year = '01-01-2014' then 'Population Count' else 0 end) as population_2014
444 from edu_inc
445 group by 'Educational Attainment'
446 having sum(case when Year = '01-01-2014' then 'Population Count' else 0 end) > (
447     select avg(total_population)
448     from (
449         select Year, 'Educational Attainment',
450             sum('Population Count') as total_population
451         from edu_inc
452         group by Year, 'Educational Attainment'
453     ) a
454     where a.'Educational Attainment' = edu_inc.'Educational Attainment'
455 );

```

Educational Attainment	population_2014
High school or equivalent	6323174
Some college, less than 4-yr degree	7781293
Bachelor's degree or higher	8528169

Result 39 x

Output

California Education & Income, 2008–2014

What seven years of population data reveal, by age, gender, education, and personal income.

+6.3%

Steady population growth

Total population rose from 26.5M to 28.2M — dipping only in 2009, then growing every year through 2014.

32K

Gender gap nearly closed

The male–female population gap shrank from 265K in 2008 to just 32K by 2014.

+1.16M

Education paid off

Bachelor's degree holders grew the most of any group; the no-diploma population was the only category to shrink (–381K).

18.4%

More Californians report no income

The "No Income" bracket's share of the population rose from 16.4% (2008) to 18.4% (2014).

+20%

Population is aging

The 65–80+ age group grew fastest of all three age bands; the under-18 population slightly declined.

+172K

Top-gaining segment

Working-age (18–64) adults with a bachelor's degree earning \$75,000+ saw the single largest population gains, in both genders.

Highly-educated, high-earning adults gained the most people

Age	Gender	Education	Income	2008	2014	Change
18 to 64	Female	Bachelor's degree or higher	\$75,000 and over	781,211	953,282	+172,071
18 to 64	Male	Bachelor's degree or higher	\$75,000 and over	1,456,566	1,628,605	+172,039
18 to 64	Female	High school or equivalent	No Income	528,876	670,294	+141,418
18 to 64	Male	Some college, <4-yr degree	\$15,000–\$24,999	416,638	556,174	+139,536
18 to 64	Female	Bachelor's degree or higher	No Income	224,757	359,264	+134,507

Reading the pattern

- Four of the top five gaining segments are working-age adults (18–64) with a Bachelor's degree or higher.
- The two largest gains — women and men earning \$75,000+ with a bachelor's degree — grew by almost identical amounts, suggesting the recovery years disproportionately benefited highly-educated, high-earning Californians.

SQL methodology pillars

Executing advanced relational operations across four modular technical stages.

1

Self-Joins · Questions 1–6

Cross-year comparisons (2008 vs 2014) and demographic segment growth isolation.

2

Window Functions · Questions 7–16

RANK, DENSE_RANK, ROW_NUMBER, and LAG for temporal rankings and growth percentages.

3

CTEs · Questions 17–22

Modularizing multi-step logic for population classifications and consistency checks.

4

Subqueries · Questions 23–28

Dynamic benchmarks, average thresholds, and multi-layered filtering conditions.

Conclusion

Actionable socioeconomic impact

This project successfully connects demographic population counts, educational attainment milestones, and personal income distribution into a unified, high-performance MySQL 8.0 analytical framework ready for executive deployment.

Thank you!

California Educational Attainment & Personal Income (2008–2014) — SQL Project

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